Follow-Up Backtest Analysis:
Optimized RSI Strategy on the FX Market

Carnegie Mellon University
Quantitative Financial Analytics and Algorithmic Trading
Course ID: 36-680

Pei-An (Perry) LIn

Table Of Contents

Table Of Contents **1**
Strategies Overview **2**
Background on RSI (Relative Strength Index) 2
2-hour RSI Strategy 2
Strategy Description **3**
Indicators 3
Position Sizing 3
Opening a Position (Buy Signal & Take-Profit) 3
Closing a Position (Sell Signal & Stop-Loss) 3
Backtesting **4**
Initial Optimized Strategy Performance 4
RSI Strategy on FX Market Performance 5
Performance Metrics Comparison 5
Equity Plot Comparison 5
References **7**
Appendices **8**

Strategies Overview

The RSI Strategy is a mean-reversion trading approach designed for the foreign exchange (FX) market. This strategy utilizes the Relative Strength Index (RSI) to identify extreme market conditions and potential price reversals. Specifically, the strategy calculates the RSI over a designated period using hourly data. The following section discusses the background of the RSI indicator, including its founder and its measurement methodology.

Background on RSI (Relative Strength Index)

The Relative Strength Index (RSI) was developed by J. Welles Wilder Jr., a mechanical engineer turned technical analyst. He introduced the RSI in his seminal 1978 book, "New Concepts in Technical Trading Systems." Wilder's work created the foundation for many modern technical indicators, including the RSI. The RSI is a momentum oscillator that measures the speed and change of price movements, oscillating between 0 and 100. It is commonly used to identify overbought or oversold conditions. Overbought conditions are typically indicated when the RSI is above 70, suggesting that a price correction or reversal may occur. Oversold conditions are indicated when the RSI is below 30, signaling a potential upward reversal.

2-hour RSI Strategy

The 2-hour RSI strategy is a shorter-term variant of the standard RSI strategy, designed to capture quick market reversals. In the initial backtest, a 2-day RSI period was used, but for this analysis was later adjusted from 48 hours to 2 hours to align with newly adopted hourly timestamp data while preserving the strategy's short-term focus. This adjustment results in a more responsive approach, aimed at identifying rapid price movements within hourly timeframes.

Strategy Description

In the initial backtest report, the RSI strategy employed a 2-day RSI period, categorizing it as a shorter-term variant. Utilizing hourly timestamp data, the RSI period was adjusted from 48 hours to 2 hours to maintain a similar short-term approach. This adjustment transitions the strategy from a 2-day RSI to a 2-hour RSI while ensuring that all other parameters remain consistent with those optimized during the initial backtest analysis. By applying these optimized parameters to selected fiat currency pairs in the FX market, the study evaluates the strategy's performance across various markets and different fiat pairs.

Indicators

- Relative Strength Index (RSI)::
  - **Original:** 2-day RSI (48 periods for hourly data)
  - **Updated:** 2-hour RSI (2 periods for hourly data)
  - Thresholds
    - Oversold: 20
    - Overbought: 90

Position Sizing

- 2% risk-based position sizing for both buying and selling

Opening a Position (Buy Signal & Take-Profit)

- A buy signal is generated when the 2-hour RSI crosses below 20.
- This signal suggests that the price has dropped significantly and may be set for a reversal.
- 10% take-profit

Closing a Position (Sell Signal & Stop-Loss)

- A sell signal occurs when the 2-hour RSI crosses above 80, indicating the asset is overbought and a price correction is likely.
- This signal suggests it’s time to exit the position as the price may be reaching a peak.

Backtesting

Initial Optimized Strategy Performance

Appendix A displays the performance metrics tables for the optimized, original, and reproduced strategies. **Appendix B** contains the backtest plot for the optimized strategy.

The Optimized RSI Strategy began with an initial equity of $100,000 and increased to a final equity of $1,335,480.11, resulting in a total return of 1235.48% and a Compound Annual Growth Rate (CAGR) of 9.76%. The strategy achieved a Sharpe Ratio of 0.65, reflecting strong risk-adjusted returns, and experienced a maximum drawdown of -38.27%, indicating significant volatility during certain periods. With 626 total trades and being in the market 36.64% of the time, the Optimized RSI Strategy demonstrated steady growth, enduring notable declines during market crises such as those in 2008 and 2020, yet consistently rebounding thereafter. These performance metrics highlight the strategy's effectiveness and resilience across various fiat currency pairs in the FX market.

RSI Strategy on FX Market Performance

In the following backtest, a set of eight fiat currencies is used to evaluate the performance of the optimized RSI strategy in the foreign exchange (FX) market. These currencies include the USD (U.S. Dollar), EUR (Euro), GBP (British Pound), JPY (Japanese Yen), CHF (Swiss Franc), CAD (Canadian Dollar), AUD (Australian Dollar), and NZD (New Zealand Dollar).

Performance Metrics Comparison

Please refer to **Appendix C** for the performance metrics table for all fiat currency pairs.

The top three performing currency pairs in terms of final equity from the RSI strategy on the FX market were AUD/USD, AUD/CHF, and JPY/CHF. Below is a summary of their performance metrics:

- **AUD/USD** demonstrated the strongest performance with a final equity of $103,615.39, reflecting a total return of 3.615% and a CAGR of 26.23%. Despite a relatively low maximum drawdown of -1.922%, this pair achieved a Sharpe Ratio of 3.799, suggesting a favorable risk-adjusted return. It executed 83 trades and was in the market for 59.00% of the time.
- **AUD/CHF** followed closely with a final equity of $103,222.58 and a total return of 3.223%. Its CAGR reached 22.32%, but it experienced a higher maximum drawdown of -3.338%. The Sharpe Ratio of 2.150 reflects moderate risk-adjusted performance, with 85 trades executed and 60.40% of the time spent in the market.
- **JPY/CHF** ranked third with a final equity of $103,110.59 and a total return of 3.111%. It achieved a CAGR of 21.48% with a maximum drawdown of -1.903%, reflecting controlled volatility. Its Sharpe Ratio stood at 1.982, and it executed 77 trades, spending 63.55% of the time in the market.

Equity Plot Comparison

Please refer to **Appendix D, E,** and **F** for the backtest plot for the top three performing currency pairs in terms of final equity from the RSI strategy on the FX market

- **AUD/EUR:** The equity curve for the AUD/EUR pair shows minimal outperformance by the RSI strategy compared to the Buy-and-Hold approach. Both strategies closely tracked each other throughout the period, with only a slight edge in favor of the RSI strategy towards the end. The final equity difference is small, indicating that the RSI strategy captured only marginally better returns, resulting in an equity of just over $103,000, only slightly above the Buy-and-Hold.
- **AUD/CHF:** For the AUD/CHF pair, the equity curve shows more significant growth compared to the Buy-and-Hold strategy. The RSI strategy equity consistently outperforms Buy-and-Hold, with the final equity exceeding $106,000. This suggests that the RSI strategy was able to capture better trading opportunities during the period, resulting in higher returns
- **JPY/CHF:** The equity curve for the JPY/CHF pair demonstrates a relatively stable performance, with both the RSI strategy and the Buy-and-Hold approach following a similar path for most of the period. The RSI strategy gradually edges ahead towards the end, achieving a final equity of

just over $103,000, highlighting its ability to generate slightly higher returns than the Buy-and-Hold strategy

In summary, while the AUD/CHF pair showed the most notable outperformance by the RSI strategy, AUD/EUR only managed to secure a small advantage, with JPY/CHF showing a moderate edge towards the end.

References

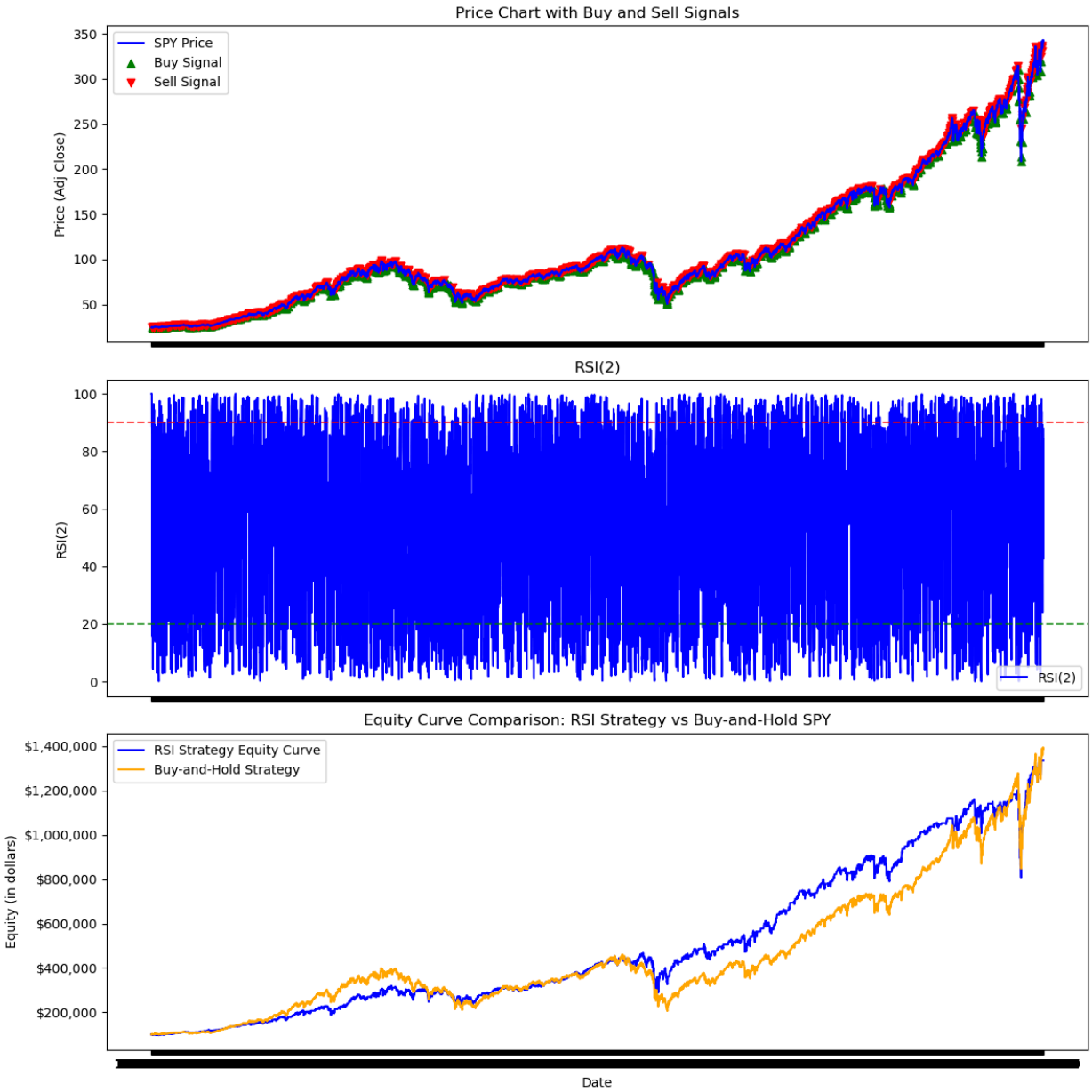
Groette, O. (2020). RSI trading strategy. Quantified Strategies. https://www.quantifiedstrategies.com/rsi-trading-strategy/

Appendices

Appendix A: Performance Metrics Comparison Optimized Strategy

Metric	Original Strategy	Reproduced Strategy	Optimized Strategy
Initial Equity	$100,000	$100,000	$100,000
Final Equity	$861,655	$861.433.64	$1,335,480.11
Sharpe Ratio	NA	0.58	0.65
Max Drawdown (%)	-33%	-36.31%	-38.27%
Total Trades	NA	658	626
Total Returns	761.66%	761.43%	1235.48%
CAGR	8.3%	8.05%	9.76%
Time in Market (%)	42%	39.01%	36.64

Appendix B: Initial Optimized Backtest Plot



Appendix C: Performance Metrics Table Sorted By Final Equity(All Fiat Currency Pairs)

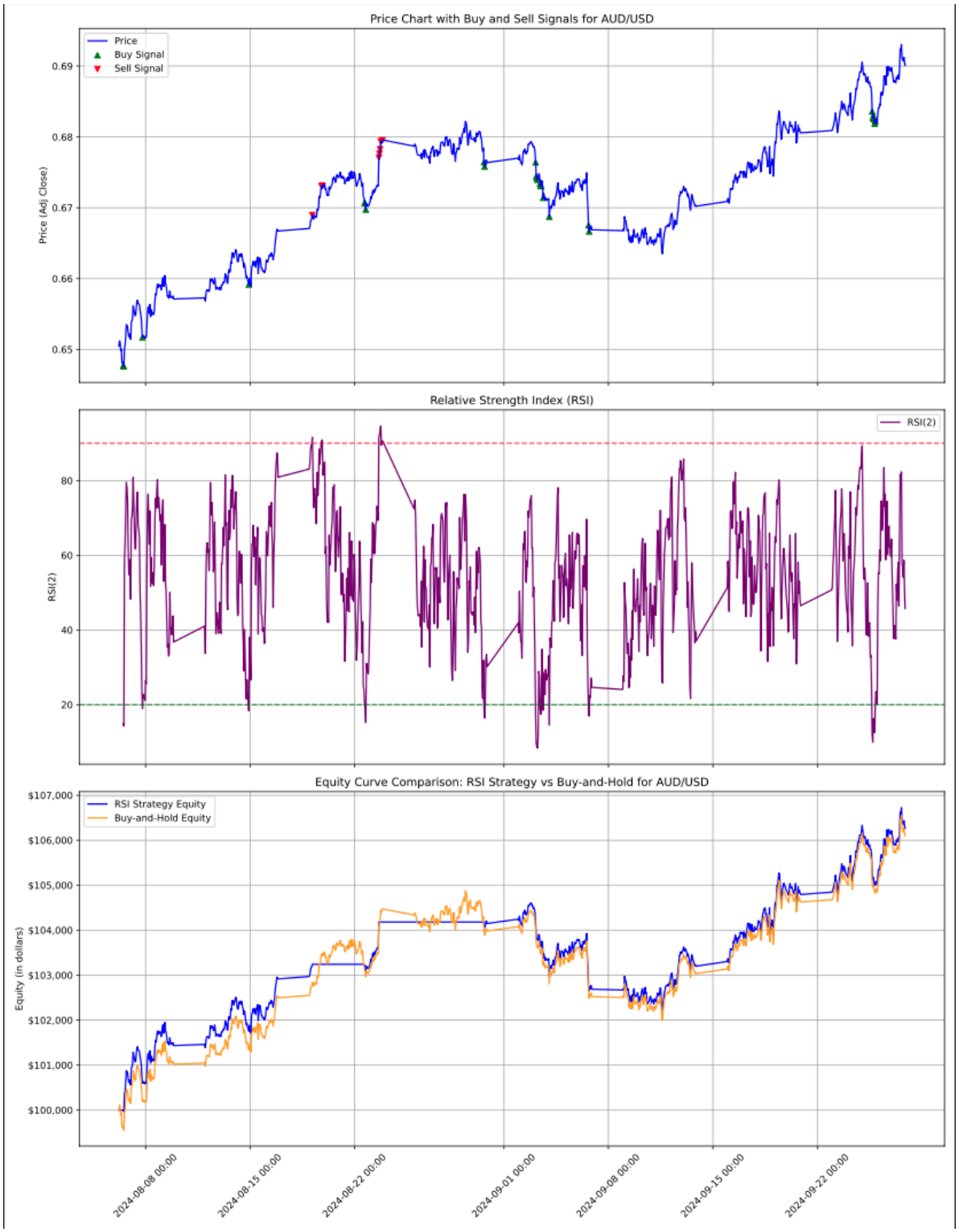
Currency Pair	Final Equity	Total Return (%)	CAGR (%)	Max Drawdown (%)	Sharpe Ratio	Total Trades	Time in Market (%)
AUD/USD	103615.393	3.615	26.234	-1.922	3.799	83	59.002
AUD/CHF	103222.576	3.223	22.323	-3.338	2.150	85	60.399
JPY/CHF	103110.590	3.111	21.483	-1.903	1.982	77	63.550
NZD/CHF	103085.171	3.085	21.317	-1.850	2.421	95	63.091
NZD/USD	102042.497	2.042	14.117	-1.833	2.048	85	57.883
AUD/EUR	102018.210	2.018	13.356	-2.418	1.893	91	63.278
JPY/GBP	101983.420	1.983	12.999	-4.270	1.040	92	63.580
GBP/USD	101638.321	1.638	11.096	-1.059	2.539	79	51.713
NZD/JPY	101629.382	1.629	10.649	-5.146	0.911	95	60.248
AUD/JPY	101546.190	1.546	10.072	-5.586	0.788	95	62.978
AUD/CAD	101539.684	1.540	10.518	-1.301	2.126	85	56.818
EUR/USD	101297.881	1.298	8.445	-1.344	1.751	83	59.667
AUD/GBP	101293.895	1.294	8.769	-1.864	1.569	76	66.811
JPY/USD	101106.410	1.106	7.312	-2.519	0.667	84	60.657
JPY/EUR	100998.724	0.999	6.426	-3.293	0.624	86	59.689
CAD/USD	100805.563	0.806	5.515	-1.254	1.372	75	62.942
GBP/CAD	100779.655	0.780	5.175	-1.317	1.469	79	58.432
GBP/EUR	100585.815	0.586	3.717	-1.469	0.956	79	65.806
GBP/CHF	100565.015	0.565	3.587	-2.433	0.562	85	61.117
NZD/EUR	100190.287	0.190	1.197	-1.527	0.233	69	59.627
CHF/USD	100061.432	0.061	0.395	-1.905	0.092	74	63.868
EUR/JPY	100055.928	0.056	0.351	-3.083	0.085	89	62.759
CHF/JPY	99969.100	-0.031	-0.193	-3.446	0.015	89	64.809
NZD/AUD	99521.945	-0.478	-3.088	-1.612	-0.686	82	62.554
NZD/CAD	99346.354	-0.654	-4.202	-2.451	-0.797	73	58.442
NZD/GBP	99315.920	-0.684	-4.366	-1.916	-0.894	78	61.828
CHF/EUR	98781.160	-1.219	-7.383	-3.106	-1.352	86	60.600

AUD/NZD	98528.534	-1.471	-9.247	-2.423	-2.241	77	69.156
USD/CAD	98525.887	-1.474	-9.379	-2.258	-2.797	76	71.382
CHF/GBP	98520.860	-1.479	-8.882	-2.546	-1.454	78	64.190
GBP/AUD	98466.113	-1.534	-9.623	-1.806	-2.049	75	65.584
GBP/JPY	98430.333	-1.570	-9.375	-4.705	-0.825	91	60.802
EUR/CAD	98325.972	-1.674	-10.130	-2.790	-2.570	72	64.226
JPY/AUD	98115.437	-1.885	-11.219	-5.480	-0.845	92	64.529
GBP/NZD	98035.033	-1.965	-12.144	-2.262	-2.417	81	62.999
CAD/EUR	97739.227	-2.261	-13.365	-2.577	-3.747	75	65.664
EUR/CHF	97663.598	-2.336	-13.771	-2.336	-2.828	75	66.218
EUR/GBP	97554.989	-2.445	-14.357	-2.894	-4.448	70	74.327
CAD/GBP	97317.141	-2.683	-16.194	-3.532	-4.535	71	66.273
JPY/CAD	96891.456	-3.109	-17.939	-5.123	-1.568	82	57.246
CAD/CHF	96857.879	-3.142	-18.219	-3.344	-3.165	73	66.771
CHF/AUD	96846.041	-3.154	-18.265	-4.503	-2.309	76	65.453
CAD/AUD	96727.307	-3.273	-19.571	-3.835	-4.390	71	71.861
CHF/CAD	96472.928	-3.527	-20.134	-4.503	-3.439	66	63.251
JPY/NZD	96352.833	-3.647	-20.735	-7.670	-1.767	82	66.081
USD/EUR	96287.162	-3.713	-21.034	-4.035	-4.340	71	73.065
USD/JPY	96073.439	-3.927	-22.061	-7.874	-2.151	87	62.757
EUR/AUD	95739.217	-4.261	-23.926	-4.854	-4.771	73	69.782
CAD/JPY	95506.772	-4.493	-25.011	-7.245	-2.725	89	63.043
CHF/NZD	95331.326	-4.669	-25.892	-5.467	-3.817	76	64.974
CAD/NZD	95257.435	-4.743	-27.242	-4.873	-4.982	63	70.346
EUR/NZD	95017.314	-4.983	-27.628	-5.858	-5.250	77	68.096
USD/NZD	94851.190	-5.149	-29.249	-5.510	-4.350	75	71.104
USD/AUD	94810.929	-5.189	-29.498	-5.983	-4.807	75	72.777
USD/GBP	94725.411	-5.275	-29.384	-5.850	-6.456	63	70.276
USD/CHF	94238.763	-5.761	-31.004	-5.782	-5.522	69	67.425

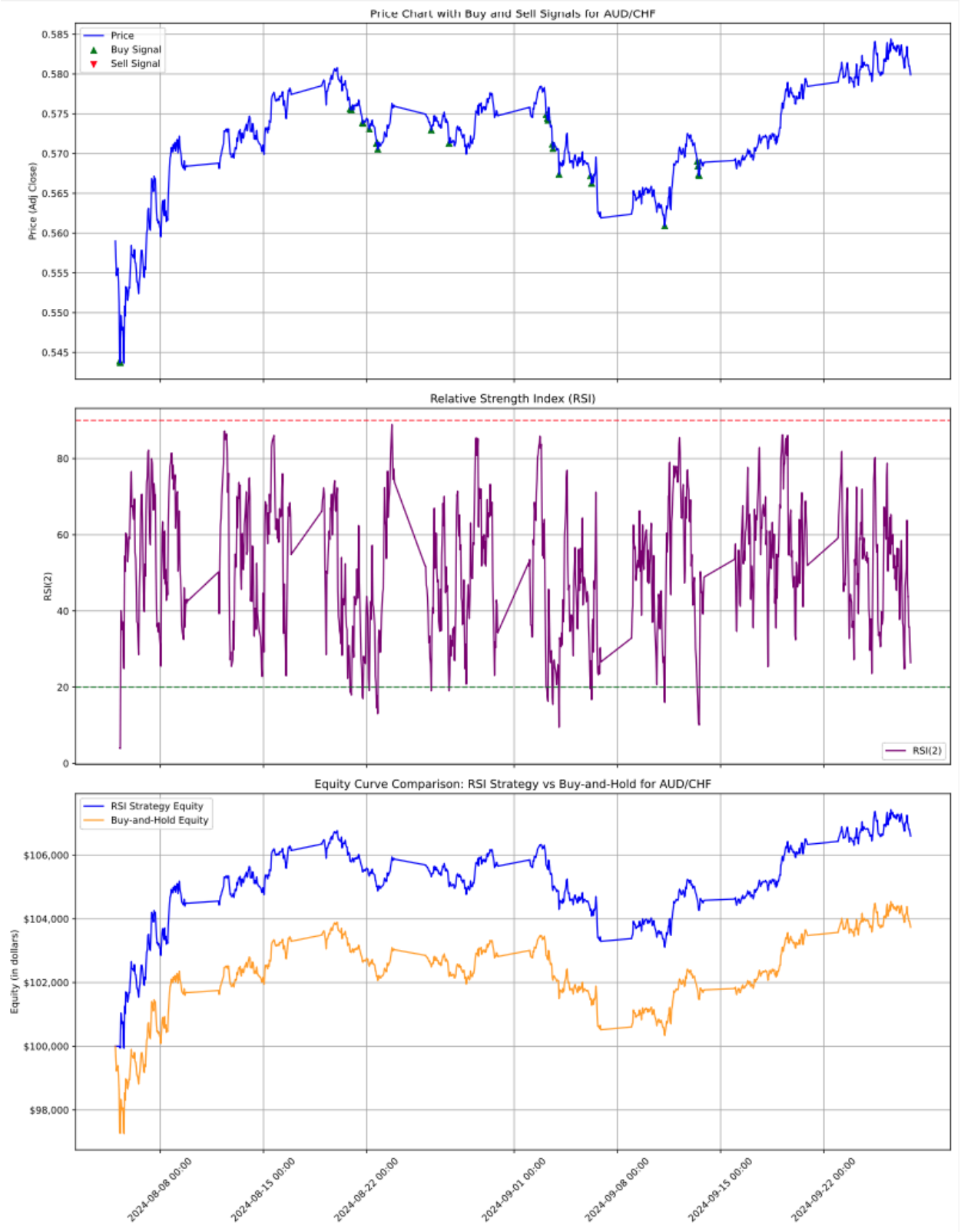
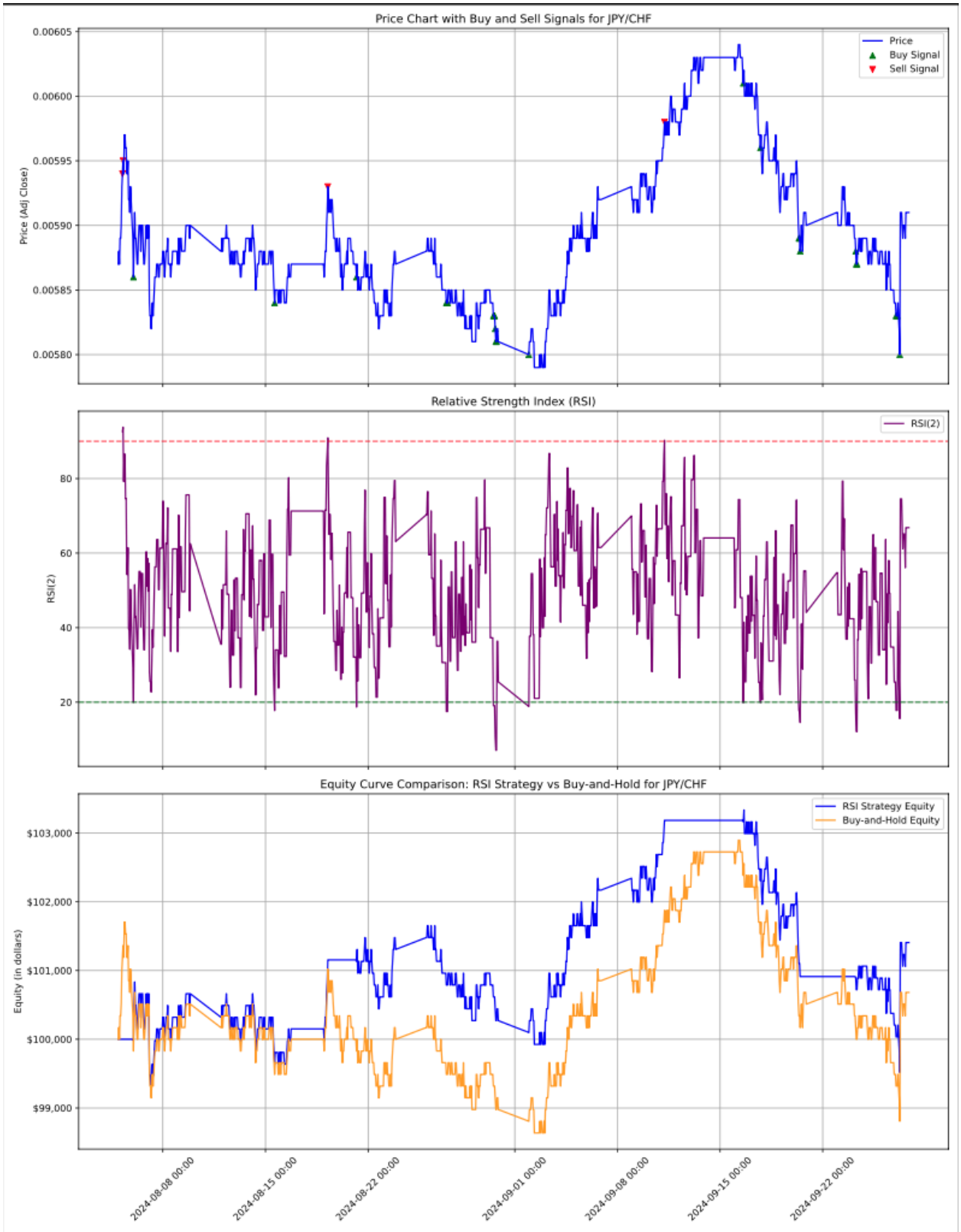
Currency Pair	Final Equity	Total Return (%)	CAGR (%)	Max Drawdown (%)	Sharpe Ratio	Total Trades	Time in Market (%)
AUD/CHF	106600.80	6.60	50.09	-3.42	3.74	1	99.26
NZD/CHF	106372.99	6.37	48.13	-2.62	4.00	1	99.26
AUD/USD	106260.64	6.26	48.93	-2.33	5.19	5	77.11
NZD/USD	106003.62	6.00	46.34	-1.93	4.88	5	68.79
GBP/USD	104597.38	4.60	33.78	-1.55	4.98	5	84.26
NZD/JPY	104401.04	4.40	30.95	-4.77	1.94	9	86.54
NZD/CAD	104075.47	4.08	29.88	-1.79	4.53	4	73.59
AUD/CAD	103955.80	3.96	28.91	-1.85	4.17	1	99.03
JPY/USD	103123.43	3.12	21.81	-3.58	1.62	7	74.34
CHF/USD	103087.87	3.09	21.59	-1.69	2.64	1	91.50
AUD/EUR	102719.85	2.72	18.34	-1.93	2.28	3	83.51
EUR/USD	102666.03	2.67	17.99	-1.67	3.01	3	93.66
NZD/EUR	102601.58	2.60	17.44	-1.63	2.46	5	69.15
JPY/CAD	102188.90	2.19	14.52	-4.09	1.15	9	72.67
GBP/CHF	102166.85	2.17	14.35	-2.38	1.71	5	88.52
AUD/JPY	102115.76	2.12	13.99	-6.36	0.91	9	83.87
GBP/JPY	101858.11	1.86	12.14	-4.65	0.90	7	87.24
JPY/EUR	101663.30	1.66	10.89	-4.70	0.90	5	89.95
CHF/CAD	101601.72	1.60	10.46	-2.31	1.52	9	64.70
CAD/CHF	101570.02	1.57	10.31	-2.56	1.26	1	98.54
JPY/CHF	101406.74	1.41	9.28	-3.69	0.84	7	79.62
EUR/CHF	101267.00	1.27	8.21	-2.74	1.25	3	93.06
GBP/EUR	100793.43	0.79	5.06	-1.43	1.41	5	38.74
GBP/CAD	100783.14	0.78	5.20	-1.25	1.30	5	67.88
NZD/AUD	100727.34	0.73	4.86	-1.33	1.07	3	83.98
CAD/JPY	100680.41	0.68	4.34	-4.61	0.37	9	93.89
AUD/GBP	100672.76	0.67	4.48	-2.13	0.71	1	95.24

AUD/NZD	100503.54	0.50	3.34	-1.42	0.73	3	83.44
NZD/GBP	100459.93	0.46	3.03	-1.06	0.67	4	55.05
CAD/USD	100439.98	0.44	2.98	-0.65	0.94	3	36.95
EUR/CAD	100286.32	0.29	1.83	-1.76	0.47	3	78.03
CAD/EUR	100285.48	0.29	1.80	-1.39	0.46	7	66.70
CHF/EUR	99768.83	-0.23	-1.44	-2.48	-0.22	5	85.63
CHF/AUD	99602.72	-0.40	-2.47	-2.58	-0.30	7	70.86
CHF/JPY	99588.28	-0.41	-2.54	-4.19	-0.27	2	77.71
GBP/NZD	98953.67	-1.05	-6.63	-1.73	-1.09	3	90.83
GBP/AUD	98876.82	-1.12	-7.13	-1.54	-1.36	3	81.60
CHF/GBP	98719.93	-1.28	-7.73	-2.57	-1.25	3	81.32
EUR/JPY	98534.69	-1.47	-8.85	-4.58	-0.69	9	77.49
JPY/AUD	98211.32	-1.79	-10.67	-5.81	-0.85	8	86.14
CAD/GBP	97781.04	-2.22	-13.56	-2.46	-3.25	5	81.95
USD/CAD	97767.70	-2.23	-13.90	-2.99	-5.00	3	51.32
EUR/GBP	97728.41	-2.27	-13.40	-3.13	-3.98	5	85.82
JPY/NZD	97480.44	-2.52	-14.75	-5.94	-1.33	8	86.66
USD/JPY	97368.40	-2.63	-15.29	-6.12	-1.08	7	94.55
JPY/GBP	97228.27	-2.77	-16.05	-5.91	-1.38	6	82.82
USD/EUR	97015.59	-2.98	-17.23	-3.31	-3.08	3	99.17
EUR/NZD	96740.45	-3.26	-18.91	-3.84	-3.35	5	88.49
EUR/AUD	96522.34	-3.48	-19.93	-4.22	-3.98	6	75.80
CAD/AUD	96476.69	-3.52	-20.93	-4.51	-4.82	5	71.10
CHF/NZD	96265.27	-3.73	-21.22	-4.40	-2.97	5	88.60
USD/GBP	96210.08	-3.79	-21.97	-5.14	-4.01	5	96.18
CAD/NZD	96208.85	-3.79	-22.35	-4.29	-3.59	3	97.73
USD/CHF	94629.56	-5.37	-29.20	-5.37	-4.91	5	77.35
USD/NZD	94434.22	-5.57	-31.26	-6.13	-4.17	5	97.62
USD/AUD	94432.08	-5.57	-31.33	-6.69	-5.51	9	68.00

Appendix D: Backtest Plot (AUD_USD)



Appendix E: Backtest Plot (AUD_CHF)



Appendix F: Backtest Plot (JPY_CHF)